

Day 19

Karnish N

(02/09/2025)

Task :

Complete process of containerizing and deploying a Web API using Docker

OUTPUT:

```
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

C:\Users\karnish.n>ssh karnish.n@192.168.7.81
(karnish.n@192.168.7.81) Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-29-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Sep  2 16:47:01 2025 from 192.168.7.64
karnish.n@dockervm01:~$ sudo docker ps -a
[sudo] password for karnish.n:
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
81bd077be52c   api_movies-api "dotnet API_Movies.d..." 56 seconds ago Up 55 seconds 0.0.0.0:8100->5000/tcp, [::]:8100->5000/tcp api-container
5feeda7e94d1   my_docker_image "dotnet Day1Zapi.dll"     About a minute ago Up About a minute 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp wonderful_rubin

karnish.n@dockervm01:~$ docker build -t petapi .
ERROR: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied
karnish.n@dockervm01:~$ sudo docker build -t petapi .
[+] Building 0.1s (1/1) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 2B                                0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory
karnish.n@dockervm01:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  thinclient_drives  Videos
karnish.n@dockervm01:~$ cd Documents
karnish.n@dockervm01:~/Documents$ cd Pet_API
karnish.n@dockervm01:~/Documents/Pet_API$ cd Pet_API
karnish.n@dockervm01:~/Documents/Pet_API/Pet_API$ docker build -t petapi .
ERROR: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied

karnish.n@dockervm01:~/Documents/Pet_API/Pet_API$ sudo docker build -t petapi .
[+] Building 26.9s (15/15) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 366B                                0.0s
=> [internal] load metadata for mcr.microsoft.com/dotnet/aspnet:8.0 0.2s
=> [internal] load metadata for mcr.microsoft.com/dotnet/sdk:8.0 0.1s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                      0.0s
=> [build 1/6] FROM mcr.microsoft.com/dotnet/sdk:8.0@sha256:7fd287cec03e027e39be13fb06ca1347f1338c7a6ab4ad305d94d91852693f70 0.0s
=> [internal] load build context                                  0.2s
=> => transferring context: 42.29MB                                0.2s
=> [runtime 1/3] FROM mcr.microsoft.com/dotnet/aspnet:8.0@sha256:6d40ae71165cb8a4427477ae3b447b05c080330cabae08b0dae672552bcd34d8 0.0s
=> CACHED [build 2/6] WORKDIR /app                               0.0s
=> [build 3/6] COPY *.csproj ./                                  0.0s
=> [build 4/6] RUN dotnet restore                                  20.3s
=> [build 5/6] COPY . .                                           0.2s
=> [build 6/6] RUN dotnet publish -c Release -o out               5.5s
=> CACHED [runtime 2/3] WORKDIR /app                               0.0s
=> [runtime 3/3] COPY --from=build /app/out ./                     0.1s
=> exporting to image                                              0.1s
=> => exporting layers                                             0.1s
=> => writing image sha256:f7505aa5bcd65358466cb12051c600503095ac712349d86b564a2b46f2527fdd 0.1s
=> => naming to docker.io/library/petapi                          0.0s
karnish.n@dockervm01:~/Documents/Pet_API/Pet_API$ sudo docker ps -aq --filter ancestor=my-dotnet | xargs docker rm -fsudo docker ps -a # find container ID
unknown shorthand flag: 's' in -sudo
Usage:  docker rm [OPTIONS] CONTAINER [CONTAINER...]

Run 'docker rm --help' for more information.
karnish.n@dockervm01:~/Documents/Pet_API/Pet_API$ sudo docker rm <container_id>
-bash: syntax error near unexpected token `newline'
karnish.n@dockervm01:~/Documents/Pet_API/Pet_API$ sudo docker run -d -p 8087:8080 petapi
d04a53f36a663a8195728e7cadc861acc5c43f51b7687cc2bcccda8ab62fc3ed
karnish.n@dockervm01:~/Documents/Pet_API/Pet_API$
```